

SESSION 6 • LECTURE 5

Temperature Scales

Detailed lecture notes — the four scales, ideal-gas absolute scale, conversions, solved numericals

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Learning objectives. After this session you should be able to: (i) relate the Celsius, Fahrenheit, Kelvin and Rankine scales; (ii) convert temperatures and temperature differences correctly; (iii) explain the constant-volume gas thermometer and the ideal-gas (absolute) scale; (iv) solve arbitrary-scale problems by fitting a straight line through two fixed points.

1. The Four Temperature Scales

Feature	Celsius (°C)	Fahrenheit (°F)	Kelvin (K)	Rankine (°R)
Type	Relative	Relative	Absolute (SI)	Absolute (English)
Ice point	0	32	273.15	491.67
Steam point	100	212	373.15	671.67
Absolute zero	−273.15	−459.67	0	0
Divisions ice→steam	100	180	100	180

$$T(K) = t(^{\circ}C) + 273.15 \quad | \quad t(^{\circ}F) = 1.8 t(^{\circ}C) + 32 \quad | \quad T(^{\circ}R) = t(^{\circ}F) + 459.67 \quad | \quad T(^{\circ}R) = 1.8 T(K)$$

KEY POINT A temperature *difference* is identical in °C and K (and identical in °F and °R): $\Delta T(K) = \Delta t(^{\circ}C)$. Never add 273 to a difference. Every thermodynamic formula ($pV = mRT$, Carnot efficiency, entropy) requires absolute temperature in kelvin.

2. The Ideal-gas (Absolute) Temperature Scale

The **constant-volume gas thermometer** holds a fixed mass of gas at constant volume and reads its pressure. With 273.16 K assigned to the triple point of water:

$$T = 273.16 \times (p / p_{tp}) \quad \text{at constant volume}$$

p = pressure at unknown T ; p_{tp} = pressure at the triple point

- Different gases give slightly different readings at ordinary pressures because no real gas is ideal.
- As the gas amount is reduced ($p_{tp} \rightarrow 0$), all gases converge to the **same** temperature — the ideal-gas scale, independent of the gas used.
- The ideal-gas scale coincides with the thermodynamic (Kelvin) scale defined from the second law — hence “absolute”.
- Absolute zero (0 K = −273.15 °C) is the temperature at which an ideal gas would exert zero pressure; it is unattainable in practice.

3. Worked Examples

WORKED EXAMPLE 5.1 — Chain conversion

Convert normal body temperature, 98.6 °F, to °C, K and °R.

Solution.

- $t(^{\circ}C) = (98.6 - 32)/1.8 = 66.6/1.8 = 37.0 \text{ }^{\circ}C$.
- $T(K) = 37.0 + 273.15 = 310.15 \text{ K}$.
- $T(^{\circ}R) = 98.6 + 459.67 = 558.27 \text{ }^{\circ}R$ (check: $1.8 \times 310.15 = 558.27 \checkmark$).

ANSWER 37 °C = 310.15 K = 558.27 °R

WORKED EXAMPLE 5.2 — Temperature differences

During heating, the temperature of a body rises by 45°C . Express this rise in K, $^{\circ}\text{F}$ and $^{\circ}\text{R}$.

Solution.

1. $\Delta T(\text{K}) = \Delta t(^{\circ}\text{C}) = 45 \text{ K}$.
2. $\Delta t(^{\circ}\text{F}) = 1.8 \times 45 = 81^{\circ}\text{F}$.
3. $\Delta T(^{\circ}\text{R}) = \Delta t(^{\circ}\text{F}) = 81^{\circ}\text{R}$.

ANSWER 45 K, 81 $^{\circ}\text{F}$, 81 $^{\circ}\text{R}$

Note: Only differences scale by 1.8; the +32 and +273.15 shifts cancel.

WORKED EXAMPLE 5.3 — An arbitrary scale $^{\circ}\text{N}$

On a new scale N, the ice point is 20°N and the steam point is 220°N . (i) Find the $^{\circ}\text{N}$ reading at 60°C . (ii) At what temperature do the $^{\circ}\text{N}$ and $^{\circ}\text{C}$ readings coincide?

Solution.

1. Linear fit $t_{\text{N}} = a t_{\text{C}} + b$ through the fixed points: $b = 20$; $220 = 100a + 20 \Rightarrow a = 2$.
2. Hence $t_{\text{N}} = 2t_{\text{C}} + 20$.
3. (i) $t_{\text{N}}(60^{\circ}\text{C}) = 2 \times 60 + 20 = 140^{\circ}\text{N}$.
4. (ii) Coincidence: $t = 2t + 20 \Rightarrow t = -20^{\circ}$. (Compare: $^{\circ}\text{C}$ and $^{\circ}\text{F}$ coincide at -40° .)

ANSWER (i) 140°N (ii) -20°

WORKED EXAMPLE 5.4 — Gas thermometer

A constant-volume gas thermometer registers 54.2 kPa at the triple point of water and 71.8 kPa in a hot bath. Find the bath temperature in K and $^{\circ}\text{C}$.

Solution.

1. $T = 273.16 \times p/p_{\text{tp}} = 273.16 \times 71.8/54.2$.
2. Ratio = 1.3247; $T = 361.9 \text{ K}$.
3. $t = 361.9 - 273.15 = 88.7^{\circ}\text{C}$.

ANSWER $T \approx 361.9 \text{ K}$ (88.7°C)

4. Practice Problems (answers given)

#	Problem	Answer
P1	Convert -40°C to $^{\circ}\text{F}$, and 500 K to $^{\circ}\text{C}$ and $^{\circ}\text{R}$.	-40°F ; 226.85°C ; 900°R
P2	A scale X reads 0°X at ice point and 150°X at steam point. What is 40°C on scale X?	60°X
P3	At what temperature do the Celsius and Fahrenheit scales read the same value?	-40°
P4	A gas thermometer reads $p = 60 \text{ kPa}$ at 273.16 K . What pressure corresponds to 400 K ?	87.9 kPa

5. Summary

- $\text{K} = ^{\circ}\text{C} + 273.15$; $^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$; $^{\circ}\text{R} = ^{\circ}\text{F} + 459.67$; $^{\circ}\text{R} = 1.8 \text{ K}$.
- Differences: $\Delta\text{K} = \Delta^{\circ}\text{C}$ and $\Delta^{\circ}\text{R} = \Delta^{\circ}\text{F} = 1.8 \Delta\text{K}$.
- Ideal-gas scale: $T = 273.16 (p/p_{\text{tp}})$ at constant volume, extrapolated to zero gas pressure.
- Arbitrary-scale problems: fit $t_{\text{new}} = a t_{\text{C}} + b$ through the two fixed points.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 2 (Temperature). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.